Gatlin A

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PROFILE

Computer Science undergraduate specializing in Machine Learning, Deep Learning, and AI Engineering, with hands-on experience building custom neural networks, LLM-based agents, anomaly detection systems, and computer vision pipelines. Strong in Python and PyTorch, with experience developing AI workflows and backend systems.

EDUCATION

Karunya University, Tamil Nadu — B.Tech, Computer Science and Engineering 2024 – Present

Good Shepherd Model School, Tuticorin — Higher Secondary (Computer Science & Mathematics) 2022 – 2024

TECHNICAL SKILLS

Languages	Python, Java, C, TypeScript
Machine Learning	PyTorch, Machine Learning, Deep Learning, NLP, Neural Networks, Computer Vision, Anomaly Detection
LLM / AI	LLMs, AI Agents, Prompt Engineering, Tokenization, Embeddings, Attention Mechanisms, LLM APIs
Backend	FastAPI, Flask, Node.js, REST APIs
Databases	PostgreSQL, MySQL, Redis, Firebase
Tools	Git, GitHub, Docker, Linux, VS Code
Core CS	Data Structures & Algorithms, Object-Oriented Programming, DBMS

EXPERIENCE

Web Development Intern — D Data Technologies June 2026

- Completed a 4-week internship contributing to web application development tasks, applying programming, debugging, and API integration practices in a professional environment.

PROJECTS

Azii — Custom AI Language Model *Python, PyTorch, NLP*

- Built a language model from scratch in PyTorch: tokenization, vocabulary construction, embeddings, attention mechanisms, and full neural architecture.
- Experimented with residual connections and iterative training strategies; validated components through tensor-shape and component-level experiments.

Autonomous LLM Agent & Tool-Use Framework *Python, Qwen, Node.js, REST APIs*

- Developed an LLM agent that converts high-level goals into structured, executable actions, with planning, short- and long-term memory, JSON outputs, action validation, and failure recovery.
- Integrated a locally hosted Qwen model via an OpenAI-compatible API; built an execution-feedback loop and refined prompt constraints to reduce invalid and repeated actions.

Cynova — Adaptive Forward Validation for Anomaly Detection *Python, PyTorch, Autoencoder*

- Designed an Adaptive Forward Validation framework using future-evidence windows and persistence thresholds to distinguish transient anomalies from persistent events.
- Trained an autoencoder on ~1 year of multivariate NOAA ASOS weather data (temperature, pressure, humidity, dew point) and produced validated incident-level results.

Real-Time Footfall Detection & Tracking *Python, PyTorch, YOLO11, BoT-SORT*

- Trained a YOLO11s person-detection model on MOT17 (13,098 images, 233,585 annotated instances), achieving 0.917 precision, 0.760 recall, 0.836 mAP@50, and 0.638 mAP@50–95.
- Integrated tracking, appearance-based Re-ID, and global identity management with line-crossing logic for directional counting.

School ERP Management System *FastAPI, SQLAlchemy, MySQL, Redis, Docker*

- Built a modular ERP backend with a layered architecture (routes, schemas, repositories, services, Alembic migrations) and a Student Information module.
- Implemented JWT auth, role-based access across 11 roles, MFA/OTP, refresh tokens, rate limiting, and audit logging.

HaveNBloom — AI-Powered Full-Stack Application *React, TypeScript, Node.js, PostgreSQL*

- Built a full-stack app with AI-powered resume analysis and skill enhancement, integrating model processing with backend services and database workflows.

CERTIFICATIONS

- Intermediate Machine Learning — Kaggle (Aug 2026)
- Introduction to Modern AI — Cisco Networking Academy (Aug 2026)
- AI Fundamentals: Foundations for Understanding AI — Cisco Networking Academy (Sep 2026)
- CCNA: Enterprise Networking, Security, and Automation — Cisco Networking Academy